(*)

$$a_0\, y(n) + a_1\, y(n-1) + \cdots + a_N\, y(n-N) = b_0\, x(n) + b_1\, x(n-1) + \cdots + b_M\, x(n-M)$$

input signal $\{x(n)\}$

output signal $\{y(n)\}$

1. Given $\{x(n)\}$ how to find output $\{y(n)\}$?
2. Will (*) LCCDE represent an LTI system?
3. Will (*) LCCDE represent a causal system?

Many different systems may have the same LCCDE representation.

$x(n) = \delta(n)$

Initial Rest: $y(0) = 0$ for all $n < 0$

$h(n) = u(n)$

A Causal Accumulator

$$y(n) = \sum_{k=-\infty}^{n} x(k)$$

$$y(n-1) = \sum_{k=-\infty}^{n-1} x(k)$$

$x(n) = \delta(n)$

$y(0) = 0$ (zero aux condition)

$h(n) = -u(-n-1)$

B Anti Causal

$$y(n) = -\sum_{k=n+1}^{\infty} x(k)$$

$$y(n-1) = -\sum_{k=n}^{\infty} x(k)$$

$$y(n-1) = \sum_{k=-\infty} x(k)$$

$$y(n) - y(n-1) = x(n)$$

LCCDE

$$y(n-1) = - \sum_{k=n} x(k)$$

$$y(n) - y(n-1) = x(n)$$

LCCDE.

Solving LCCDE

(*) $a_0 y(n) + a_1 y(n-1) + \cdots + a_N y(n-N) = b_0 x(n) + \cdots + b_M x(n-M)$

degree N.

We need auxillary conditions to find the exact output signal.

- One way: we can specify the values of $\{y(-1), y(-2), \ldots, y(-N)\}$

We can solve LCCDE (*) using recursion.

Forward Recursion

$$y(n) = \frac{1}{a_0}\left[-a_1 y(n-1) - \cdots - a_N y(n-N) + b_0 x(n) + \cdots + b_M x(n-M) \right]$$

At $n=0$ we can find $y(0)$ using $\{y(-1), \ldots, y(-N)\}$

Similarly $n=1, n=2, \ldots$

For all $n \geq 0$ we can use forward recursion to find output.

Backward Recursion

$$y(n-N) = \frac{1}{a_N}\left[-a_0 y(n) - \cdots - a_{N-1} y(n-N+1) \right.$$

$+ b_0 x(n) + \cdots + b_M x(n-M) \Big]$

This can be need to find

output | $y(n)$ for all $n < -N$ |

- If auxillary conditions are zero

| $y(-1) = y(-2) = \cdots = y(-N) = 0$ |

This zero auxillary condition

leads to an LTI system.

- Initial Rest Condition

If $x(n) = 0$ for all $n < N_0$

then $y(n) = 0$ for all $n < N_0$

With initial rest condition, we get Causal LTI system.